

## AN ON-LINE ASSIGNMENT PROBLEM WITH RANDOM EFFECTIVENESS AND COSTLY INFORMATION

Nicholas G. HALL and Rakesh V. VOHRA

*Faculty of Management Sciences, The Ohio State University, 1775 College Road, Columbus, OH 43210-1399, USA*

Received August 1986

Revised July 1987

We study the on-line assignment problem, where the testing of effectiveness incurs a cost. An optimal testing policy to maximize expected net effectiveness is derived.

assignment problem \* on-line processing \* optimal information search

### 1. Introduction

The assignment problem is widely used in large-scale personnel planning. There are  $n$  jobs to be performed by  $n$  persons, each person performing exactly one of the jobs. A cost or effectiveness number  $C_{ij}$  is known for each person-job pair  $(i, j)$ . The optimal solution is that assignment which minimizes the total cost (or maximizes the total effectiveness) among all feasible assignments. The best known solution method for the assignment problem is the Hungarian method, described by Kuhn [3], which runs in  $O(n^3)$  time. That method is easy to implement, and is consequently effective in solving large assignment problems. Specialized solution methods are also discussed by Spivey and Thrall [5].

This paper concerns a version of the assignment problem in which jobs arrive in sequence, and the choice of which persons perform them must be made without knowledge of jobs which will arrive subsequently. Further, the effectiveness number  $C_{ij}$  dollars for each person-job pair is assumed to be unknown, but taken from a uniform distribution on the interval  $[0, 1]$ , independently of the corresponding numbers for all other pairs. However, this effectiveness number can be found exactly, through testing for that person-job pair. We assume that the testing of each such pair costs  $T$  dollars,  $T > 0$ . Clearly, an assignment which makes use of considerable testing is likely to be a more effective assignment, but the cost of testing will also be high. In this paper we discuss ways to maximize the expected net effectiveness, i.e., the expected total effectiveness of the assignment, less the costs of testing. This problem may be considered as a generalization of the classical 'secretary' problem (Gilbert and Mosteller [2], Rose [4]) in which  $n$  appointments, rather than one, must be made, and where testing is costly. We next describe an example application for the model described.

Consider  $n$  distinct data bases, and  $n$  blocks of data requiring storage. Each block of data is of uncertain size but within the capacity of the data base. Further, each data base has a known access time per unit of data. It is possible to examine a block of data, in order to assess its size, but time is lost in doing so. The minimization of the expected total access plus testing time is achieved by the model suggested. In general, our model is applicable to any on-line assignment problem where (a) the effectiveness of an individual person-job assignment can be considered as a uniformly distributed random variable, independent of the effectiveness of other assignments, and (b) that effectiveness can be found exactly through a costly testing procedure. In the remainder of the paper, we present and solve a mathematical model for the maximization of the expected net effectiveness of the on-line assignment problem. Then we perform sensitivity analysis for the effect of changes in the cost-of-testing parameter  $T$  on the amount of testing and the optimal value.

## 2. Formulation and solution approach

We assume that  $n$  persons await the arrival of the first job. When that job arrives, the effectiveness of each of  $t$  persons in performing that job is tested, where  $0 \leq t \leq n$ . After testing, the most effective person tested is assigned to perform the job. Then the remaining  $(n-1)$  persons await the next job, and the process repeats. This procedure runs in  $O(n^2)$  time, because for each of the  $n$  jobs there may be as many as  $n$  tests and  $(n-1)$  comparisons in determining the assignment for that job. We seek a policy which tests the same number,  $t$ , of persons at each stage, while there are at least  $t$  persons remaining unassigned. If there are fewer than  $t$  persons remaining unassigned, all those persons are tested. Among the advantages of this policy are its simplicity and ease of implementation. But an even more important advantage is its robustness. If the number of jobs exceeds or falls short of the number expected, this policy can still be used. Moreover, we prove in Lemma 1 that this policy is optimal. The optimal information search strategy is specified by the value of  $t$ ,  $0 \leq t \leq n$ , which maximizes the expected net effectiveness of the assignment.

In the analysis which follows, the arrival of each job defines an iteration. The number of person-job pairs examined by this strategy is  $t, t, \dots, t, t-1, t-2, \dots, 3, 2, 0$ , giving a total of  $(n-t+1)t + \sum_{i=1}^{t-1} i - 1$  pairs. Note that it is never worthwhile to incur the cost of examining exactly one pair and then choosing it. Assuming that at least one person has been tested for a job, that job is always assigned to the person who is most effective, among those tested, for that job. If no persons have been tested, the assignment is made randomly among the persons not so far assigned. Now we find an expression for the expected net effectiveness of the assignment, when  $t$  is the number of persons tested at the first iteration. Let  $X_1, \dots, X_t$  be independent random variables, uniformly distributed on the interval  $[0, 1]$ , and let  $X(t) = \max\{X_1, \dots, X_t\}$ . Then  $E[X(t)] = t/(t+1)$ . The objective function is defined separately for  $t=0$  and for  $2 \leq t \leq n$ . The maximization of expected net profit is equivalent to  $\max\{Z_1^*, Z_2^*\}$ , where

$$Z_1^* = \max_{t \in I_{2,n}} Z_1(t) = (n-t+1)t/(t+1) + \sum_{k=1}^{t-1} (k/k+1) - Tt(n-t+1) - Tt(t-1)/2 + T, \quad (1)$$

$$Z_2^* = n/2, \quad (2)$$

with  $I_{2,n}$  being defined as the set of integers in the interval  $[2, n]$ . In expression (1), the first term represents the expected effectiveness of assignments, while the number of persons remaining unassigned is greater than or equal to  $t$ . The second term in (1) is the expected effectiveness when all (fewer than  $t$ ) persons hitherto unassigned are tested. Similarly the third and fourth terms represent the costs of testing  $t$  persons for the first  $(n-t+1)$  iterations, and then  $t-1, t-2, \dots, 3, 2, 0$  persons at subsequent iterations. Note that (2) defines the expected net benefit for the option of doing no testing at any iteration, because the expected effectiveness of each of the  $n$  random person-job assignments is  $1/2$ .

**Lemma 1.** *The maximization of expected net effectiveness is achieved by a policy which tests the same number of persons,  $t^*$ ,  $0 \leq t^* \leq n$ , while there are at least  $t^*$  persons unassigned, and thereafter tests all those unassigned until only one remains.*

**Proof.** Consider the following simple dynamic programming model. Let  $f_n$  = the expected net effectiveness for the problem with  $n$  persons unassigned. If we test  $t$ ,  $0 \leq t \leq n$ , persons at a stage, we have the recurrence relation

$$f_n = \max_{0 \leq t \leq n} [t/(t+1) - tT] + f_{n-1},$$

where  $g(t) = t/(t+1) - tT$  is independent of  $n$ . Consequently, the same number of persons,  $t^*$ , are tested at each stage, if at least  $t^*$  persons remain unassigned. If fewer than  $t^*$  persons remain unassigned, all those persons are tested. This is because  $g''(t) = -2/(t+1)^3 < 0$  for  $t \geq 0$ , hence  $g(t)$  is unimodal.  $\square$

We describe in Section 3 how to find the value of  $t$ ,  $2 \leq t \leq n$ , which maximizes  $Z_1(t)$ .

### 3. The optimal amount of information

Since expression (1) is defined only on the integers between 2 and  $n$ , it is not differentiable. However, we can show that it is a concave function of its argument, using the following result.

#### Lemma 2

$$Z_1(t+1) - Z_1(t) \leq Z_1(t) - Z_1(t-1), \quad 2 \leq t \leq n-1.$$

**Proof.** First,

$$\begin{aligned} Z_1(t+1) - Z_1(t) &= (n-t)(t+1)/(t+2) + (t/t+1) - T(t+1)(n-t) - Tt(t+1)/2 + T \\ &\quad - (n-t+1)(t/t+1) + Tt(n-t+1) + Tt(t-1)/2 - T \\ &= T(nt - t^2 + t + t^2/2 - t/2 - t^2/2 - t/2 - nt - n + t^2 + t) \\ &\quad + (n-t)(t+1/t+2 - t/t+1) \\ &= T(t-n) + (n-t)/(t+1)(t+2) \\ &= (n-t)[-T + 1/(t+1)(t+2)]. \end{aligned} \quad (3)$$

Also,

$$\begin{aligned} Z_1(t) - Z_1(t-1) &= (n-t+1)(t/t+1) + (t-1)/t - Tt(n-t+1) - Tt(t-1)/2 + T \\ &\quad - (n-t+2)(t-1)/t + Tt(n-t+2) + T(t-1)(t-2)/2 - T \\ &= T[nt - t^2 + 2t - nt + t^2 - t + (t^2 - 3t + 2 - t^2 + t)/2] \\ &\quad + (n-t+1)[(t/t+1) - (t-1)/t] \\ &= T + (n-t+1)/t(t+1). \end{aligned} \quad (4)$$

If Lemma 2 does not hold, then (4) < (3), i.e.,

$$T + (n-t+1)/t(t+1) < (n-t)/(t+1)(t+2) - (n-t)T,$$

which gives

$$(n-t+1)T < (n-t)/(t+1)(t+2) - (n-t+1)/t(t+1) < 0,$$

which is a contradiction, since  $n-t+1 \geq 0$  and  $T \geq 0$ .  $\square$

Let  $t^*$  denote the value of  $t$ ,  $t \in I_{2,n}$ , which maximizes  $Z_1(t)$ . It is now easy to find  $t^*$ .

#### Lemma 3

$$t^* = \min_{t \in I_{2,n}} \{t \mid T \geq 1/(t+1)(t+2)\}. \quad (5)$$

**Proof.** From Lemma 2,  $Z_1(t)$  is a unimodal concave function of  $t$ . Then it follows immediately that  $t^* = \min_{t \in I_{2,n}} \{t \mid Z_1(t+1) \leq Z_1(t)\}$ , and the result follows, using (3), and the fact that  $n \geq t$ .  $\square$

From Lemma 3, it is possible to derive a closed form expression for the optimal amount of information to obtain.

#### Theorem

$$t^* = \max\left\{2, \min\left\{n, \left\lceil -3/2 + (1 + 4/T)^{1/2}/2 \right\rceil\right\}\right\}. \quad (6)$$

**Proof.** From (1), we require  $2 \leq t^* \leq n$ . From Lemma 3, we require  $T \geq 1/(t^* + 1)(t^* + 2)$ , which implies  $t^{*2} + 3t^* + 2 - 1/T \geq 0$ . Note that this quadratic equation has only positive coefficients for  $t^{*2}$  and  $t^*$ , therefore the smallest integer greater than or equal to the positive root of  $t^{*2} + 3t^* + 2 - 1/T = 0$  will satisfy (5), hence the result.  $\square$

A few comments about the theorem are in order. First, recall that the solution obtained by setting  $t^*$  according to (6) is only optimal if  $Z_1^* \geq n/2$ . Next, the interesting property of (6) is that it contains no terms in  $n$ . More precisely, for an interior solution, i.e.,  $2 \leq t^* < n$ , the optimal amount of information to obtain for the on-line assignment problem is independent of the problem size. The theorem suggests the following simple algorithm for the maximization of expected net profit.

**Algorithm**

**Step 0.** Input  $n$  (the problem size) and  $T$  (the unit cost of testing), go to Step 1.

**Step 1**

$$t^* \leftarrow \max \left\{ 2, \min \left\{ n, \left\lceil -3/2 + (1 + 4/T)^{1/2}/2 \right\rceil \right\} \right\},$$

go to Step 2.

**Step 2**

$$Z_1^* \leftarrow (n - t^* + 1)(t^*/t^* + 1) + \sum_{k=1}^{t^*-1} (k/k + 1) - Tt^*(n - t^* + 1) - Tt^*(t^* - 1)/2 + T,$$

$$Z_2^* \leftarrow n/2.$$

If  $Z_1^* < Z_2^*$ , go to Step 3. Otherwise, go to Step 4.

**Step 3.** Output: no testing to be performed, maximum expected net profit  $= n/2$ . Terminate.

**Step 4.** Output: test  $\min \{t^*, r\}$  persons for each arriving job, where  $r$  is the number of persons hitherto unassigned; also, maximum expected net profit  $= Z_1^*$ . Terminate.

The second term in  $Z_1^*$  requires comment, if  $t^*$  is very large, because there is no known exact closed form expression for its value. However,

$$\sum_{k=1}^{t^*-1} (k/k + 1) = t^* - 1 - \sum_{k=1}^{t^*-1} 1/(k + 1) = t^* - 1 - \sum_{k=1}^{t^*} 1/k + 1,$$

and we also know  $\ln t^* \leq \sum_{k=1}^{t^*-1} 1/k \leq 1 + \ln t^*$  (see Clapham [1]), therefore

$$t^* - \ln t^* - 1 \leq \sum_{k=1}^{t^*-1} (k/k + 1) \leq t^* - \ln t^*. \quad (7)$$

Taking the midpoint of the interval defined by (7) as an estimate, we obtain an estimate for  $Z_1^*$  in Step 2, with a maximum error of  $\pm 1/2$ . Clearly this error could cause an incorrect result in the comparison of  $Z_1^*$  and  $Z_2^*$  at Step 2 of the algorithm. Let  $Z^*$  denote the objective value obtained by evaluating the second term in  $Z_1^*$  exactly, and let  $Z_e$  denote the objective value obtained from the approximation suggested. Then it is easy to see that  $Z^* - Z_e \leq 1/2$ . However,  $Z^* \geq n/2$ , therefore  $(Z^* - Z_e)/Z^* \leq 1/n$  and  $\lim_{n \rightarrow \infty} (Z^* - Z_e)/Z^* = 0$ . This means that the relative error of the approximation becomes zero, as the problem size becomes large. Moreover, for most problems,  $|Z_1^* - Z_2^*| > 1/2$ , in which case the solution obtained by the algorithm is provably optimal among the class of strategies defined.

#### 4. Sensitivity analysis

We next investigate the sensitivity of the optimal objective value,  $Z(t^*)$ , to changes in the cost of testing parameter,  $T$ . We would expect that  $t^*$ , the optimal number of persons to test at each stage, would decrease as  $T$  increases. Eventually  $T$  becomes sufficiently large that  $Z_2^* > Z_1^*$ , and no testing at all is

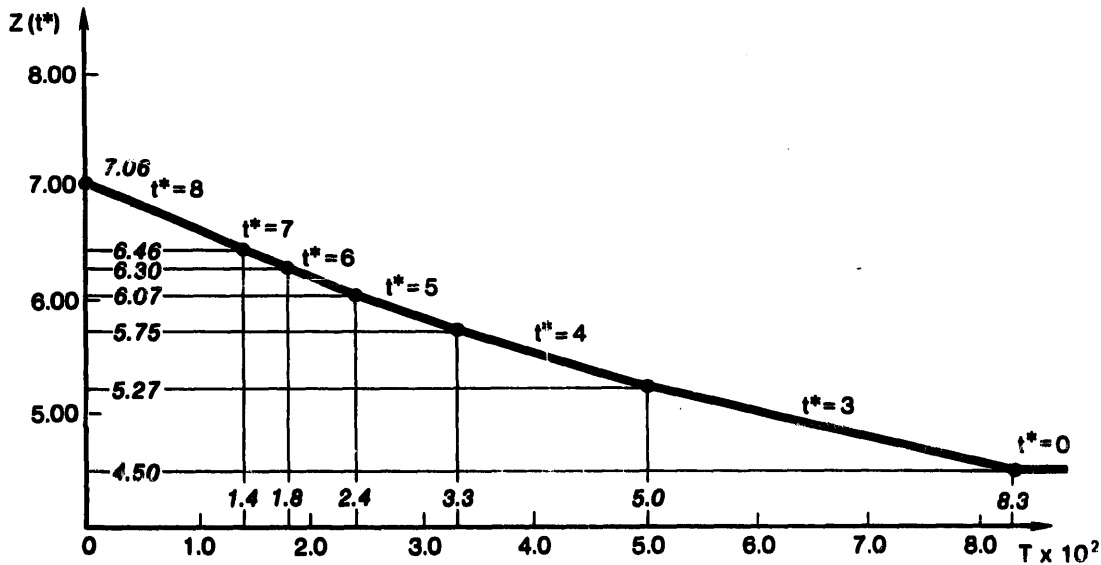


Fig. 1. Sensitivity analysis for cost of testing ( $n = 9$ ).

performed. Figure 1 demonstrates the relationship between different values of  $T$  and the optimal objective value, for a  $9 \times 9$  assignment problem. In this analysis, the evaluation of  $Z_1^*$  was performed exactly.

## 5. Conclusions and possible extensions

We have described a technique for maximizing the expected effectiveness of an assignment, less the costs of testing prior to making that assignment. This technique can be used for assignment problems which are strictly on-line in the sense that jobs arrive in sequence, and each must be assigned to a person before the next job arrives. However, an off-line assignment problem in which  $n$  jobs and  $n$  persons are initially available, can be solved in the same manner at low computational cost. Our technique is optimal for most problems, and has, in the worst case, an absolute error of  $1/2$ . This absolute error yields a relative error of zero, as the problem size becomes large. One obvious generalization of the problem described relates to the distribution from which the effectiveness of each person-job pair is assumed to come. Although the uniform distribution was used, many other bounded distributions would give a similar analysis.

## References

- [1] C.R.J. Clapham, *Introduction to Mathematical Analysis*, Routledge & Kegan Paul, London, 1973.
- [2] J.P. Gilbert and F. Mosteller, "Recognizing the maximum of a sequence", *Journal of the American Statistical Association* **61**, 35-73 (1966).
- [3] H.W. Kuhn, "The Hungarian method for the assignment problem", *Naval Research Logistics Quarterly* **2**, 83-97 (1955).
- [4] J.S. Rose, "A problem of optimal choice and assignment", *Operations Research* **30**, 172-181 (1982).
- [5] W.A. Spivey and R.M. Trall, *Linear Optimization*, Holt, Rinehart & Winston, New York, 1970.